C8 Periodicity – Knowledge Organiser

AQA A-level Chemistry • Set 1: Period 3 Trends and Structures **Paper 1**

#	Question	Answer
1	Periodicity	Repeating trends in properties across periods
2	Block containing Na and Mg	s
3	Block containing Al to Ar	p
4	Trend in atomic radius across Period 3	Decreases
5	Why atomic radius decreases across Period 3	More protons, same shielding
6	General trend in first IE across Period 3	Increases
7	Trend in electronegativity across Period 3	Increases
8	Structure of Na, Mg and Al	Giant metallic
9	Structure of Si	Giant covalent (macromolecular)
10	Structure of Ar	Monatomic

C8 Periodicity – Knowledge Organiser

AQA A-level Chemistry • Set 2: Melting Points **Paper 1**

#	Question	Answer
1	Period 3 element with highest melting point	Si
2	Period 3 element with lowest melting point	Ar
3	Formula of a phosphorus molecule	P_4
4	Formula of a sulfur molecule	S_8
5	Structure of phosphorus, sulfur and chlorine	Simple molecular
6	Why Si has a very high melting point	Many strong covalent bonds to break
7	Why Al has a higher melting point than Na	More delocalised electrons, higher ion charge
8	Why S_8 melts higher than P_4	Larger molecule, stronger van der Waals
9	Order of melting point for S_8, P_4, Cl_2, Ar	$S_8 > P_4 > Cl_2 > Ar$
10	Why Ar has the lowest melting point	Single atoms, weakest van der Waals